

UNIT 1: COMPUTER NETWORK AND COMMUNICATION

1. What is bandwidth? How is it measured?
2. What is broadband? How does it differ from dial-up connections?
3. Differentiate between LAN and MAN.
4. Differentiate between LAN and WAN.
5. What is a Personal Area Network (PAN), and how does it differ from a Local Area Network (LAN)?
6. What is a Wide Area Network (WAN)? Mention its characteristics.
7. Define a computer network. How is it useful, and how can it reduce operational costs?
8. Write any two advantages and any two limitations of a computer network.
9. What is network topology? Give any two examples.
10. Which topology is easiest to manage, and why?
11. Suggest a suitable topology for a school computer lab that can continue operating if one computer fails. Give a reason.
12. Draw and explain the ring topology, including how data flows in it.
13. How does data travel from one computer to another in a network?
14. What is transmission media? Name its two categories.
15. Differentiate between guided and unguided media with examples.
16. Which transmission medium would you use for connecting two cities? Why?
17. Which communication medium would you choose to connect two buildings of a school campus separated by a 500-meter field, and why?
18. Why is fiber optic cable faster than copper cable?
19. A company wants high-speed and secure communication. Which cable would you recommend? Why?
20. What is a data packet in networking?
21. Why is data divided into packets before being transmitted through a network?
22. What is a protocol? List any two examples.
23. What is frequency in telecommunications? Give a suitable example.
24. Mention one real-life use of satellite communication.
25. Why is wireless communication becoming more popular today?
26. How does Wi-Fi transmit data without cables?
27. A student connects a laptop to the internet at home without cables. What technology is being used, and how does it work?
28. Describe how Bluetooth communication differs from Wi-Fi in terms of range and usage.
29. Describe the RJ45 connector. Where is it commonly used?
30. How does an RJ45 connector help establish a network connection?
31. What is a media converter? Mention its main function.
32. What is the function of a repeater?
33. A signal becomes weak when traveling over a long CAT6 cable between two floors of an office. Which networking device should be used?

34. Describe the functions of a network switch.
35. Why is a switch considered an intelligent device compared to a hub?
36. Define a hub and a router, and mention one use of each.
37. Which device is needed to connect multiple Wi-Fi-enabled laptops to the internet without physical cables?
38. Differentiate between the Internet and an Intranet with any two points.
39. Which type of network—Intranet or Extranet—would be more suitable for a large multinational organization? Justify your answer.
40. A company wants to provide limited, secure access to its internal inventory database to trusted partners. Which type of network—Intranet or Extranet—should be used?
41. What is the difference between bandwidth and throughput?
42. How does higher bandwidth affect the speed and performance of data communication?
43. Differentiate between Simplex, Half-Duplex, and Full-Duplex communication modes with examples.
44. A television broadcast sends signals only from the station to viewers. Identify the communication mode and justify your answer.
45. How do the tracert and nslookup commands help in analyzing network connectivity?
46. Which command would you use to test whether a website such as www.moecdc.gov.np is reachable? Write its syntax.
47. Which command would you use in Command Prompt to view your computer's IP address and default gateway?
48. Which wireless technology is suitable for tracking books in a library or products in a warehouse automatically?

UNIT 2: DATABASE MANAGEMENT SYSTEM (DBMS)

1. Define data and information.
2. What is a database in the context of computer science? State one importance of using it.
3. Define a Database Management System (DBMS). Give two examples and state any two advantages.
4. Why is a DBMS more advantageous for managing large amounts of data than a simple spreadsheet?
5. Differentiate between a database and a DBMS.
6. Define RDBMS and SQL. Describe an RDBMS briefly.
7. Mention any two real-life applications of databases.
8. Define the terms field and record, and differentiate between them with an example.
9. Define the terms entity and attribute.
10. What is a primary key? Give a brief example and explain why it is important.
11. What is a foreign key? Explain how primary and foreign keys work together in a relational database.
12. Explain the concept of a one-to-many relationship and provide a real-world example different from the one in the textbook.
13. What is a composite key, and when might it be necessary to use one?
14. Differentiate between DDL and DML and briefly state their roles in SQL.

15. What is the function of the WHERE clause in a SELECT statement? Give one example.
16. Differentiate between the INSERT and SELECT commands.
17. Write the SQL commands to create a database named SchoolDB and to use it.
18. What is a data type in DBMS? Name any two standard SQL data types.
19. Write the basic SQL command used to insert new records into a table.
20. Write the general syntax of the SQL command used to create a new table.
21. Write the SQL command used to display all records and columns from a given table.
22. Write the SQL command used to modify or update existing records in a table.
23. Write the SQL command used to remove records from a table.
24. What are SQL constraints? Why are they important for maintaining data integrity?

UNIT 3: MULTIMEDIA

1. Define multimedia and list its five main components.
2. Explain the importance of multimedia in education and business.
3. Explain any two advantages of multimedia.
4. How does video enhance multimedia applications?
5. Explain the importance of multimedia in advertising and entertainment. Mention any two uses of multimedia in entertainment.
6. What is graphics? What is an image file format? Give three common graphics file formats.
7. Define raster graphics.
8. Differentiate between raster and vector graphics with any two points, including how they behave when resized.
9. Which type of graphic, raster or vector, would you use for designing a billboard? Justify your answer.
10. Suggest an appropriate image format for a transparent company logo. Why is PNG preferred over JPG for this purpose?
11. What is the purpose of Layers in graphics editing software?
12. Define audio and video in multimedia.
13. Explain the difference between amplitude and wavelength in an audio waveform.
14. Why does a WAV file consume more storage space than an MP3 file?
15. Explain the difference between MP3 and AAC files.
16. What is meant by resolution in multimedia?
17. Define frame rate and explain its importance.
18. Differentiate between SD and FHD/HD video resolutions.
19. Why is the MP4 video format widely used across the internet?
20. Explain how the Timeline functions in video-editing software such as CapCut.
21. Briefly explain how psychology is utilized in animation to engage viewers.
22. Define lossy and lossless compression.
23. Write any two differences between 2D and 3D animation.

24. Mention any two advantages of 3D animation over 2D animation.
25. Write a short note on Rich Text Format (RTF).
26. If you have to create a greeting card, which multimedia components would you use, and why?
27. Write short notes on Audio Waveform; Video Resolution and Framerate; Raster and Vector Graphics; and 2D and 3D Animation.

UNIT 4: PROGRAMMING IN PYTHON

1. Define data visualization in your own words and state its importance for businesses and analysts.
2. Name three popular Python libraries for data visualization.
3. What is the key characteristic of Matplotlib that offers flexibility but requires more code?
4. What is Seaborn built upon, and what type of visualizations does it primarily focus on?
5. What is a key feature of Plotly that distinguishes it from Matplotlib?
6. What is the role of `plt.xlabel()` and `plt.ylabel()` in Matplotlib?
7. What type of data is typically represented using a bar chart?
8. What type of data is best represented using a pie chart?
9. What is the purpose of the `color` argument in Plotly Express plotting functions?
10. How can data visualization help students understand exam results that are presented only in tables?
11. How can data visualization help managers make decisions from daily sales records stored in a CSV file?
12. What are the primary advantages of using Pandas for file handling in Python?
13. What is a CSV file, and why is it a common format for data exchange?
14. What is the first step needed to use Pandas in a Python script for file handling?
15. Write the basic syntax for reading a CSV file into a Pandas DataFrame.
16. How can you access a specific column after reading a CSV file into a Pandas DataFrame? Give an example.
17. Write the basic syntax for writing a Pandas DataFrame to a CSV file.
18. What happens if you try to open a non-existent file in read ('r') mode using `open()`?
19. Explain the difference between write ('w') mode and append ('a') mode when opening a file.
20. Why is it important to close a file after read or write operations?
21. Compare the `csv` module with Pandas for file handling. Why might an enterprise with millions of rows prefer Pandas?
22. Define the term module in Python.
23. What is the relationship between modules and libraries in Python?
24. Give an example of a built-in Python library and state its purpose.
25. Why are packages useful for code organization and reusability?
26. What is the purpose of using an alias when importing a module? Give an example.
27. Explain the benefit of importing one function, such as `from math import sqrt`, instead of importing the entire module.

28. Name two functions provided by the Python math library.
29. What is a runtime error? Give one example.
30. Explain the difference between an error and an exception in Python.
31. What is the role of the except block in a try-except statement?
32. When is the else block in a try-except-else statement executed?
33. What is the use of the finally block in Python exception handling?
34. Why is placing file.close() inside a finally block considered good practice in file handling?
35. What is the main benefit of using user-defined functions in programming?
36. Name the two types of Python functions.
37. Explain the difference between a parameter and an argument in Python functions.
38. What is local scope in Python? Give a brief example.
39. What is global scope in Python, and how does it differ from local scope?
40. What problems can arise when a programmer uses only global variables, and what are the advantages of local scope?
41. What is the purpose of the return keyword in a Python function?
42. What are positional arguments and how are they passed to a function?
43. What are default arguments? How do they improve the flexibility of a function call?
44. What are keyword arguments, and what advantage do they offer?
45. What happens if a void function has a return statement without any value?
46. What is the practical difference between a void function and a non-void function? When would a non-void function be required?
47. How can Python functions return multiple values simultaneously? Give a brief code example.
48. What is the primary function of input() in Python?
49. Provide an example of a string literal in Python.
50. What is an identifier in Python?
51. Define data types in Python.
52. What is the purpose of relational operators in Python? Give one example.
53. Describe the else block in an if-else statement.
54. What is a nested if statement?
55. What is the difference between a for loop and a while loop?
56. When would you typically use a for loop instead of a while loop?
57. What is the role of range() when used with for loops?
58. Define a Python list and give a simple example.
59. Discuss the concept of iteration in programming.
60. Write one use of comments in Python.
61. What is a function in Python?
62. What is the use of t.penup() in Turtle graphics?
63. What is the purpose of t.done() in a Turtle program?

64. Write a Python program to check whether an entered number is even or odd.
65. Write a Python program to calculate the square of a number.
66. Write a Python program to reverse the digits of a number.
67. Write a Python program to display the multiplication table of a number.
68. Write a Python program to find the sum of the first 10 natural numbers.
69. Write a Python program to find the sum of odd numbers from 1 to 100.
70. Write a Python program to calculate the average of three numbers.
71. Write a Python program to calculate the factorial of an entered number.
72. Write a Python program to read a radius and find the area of a circle.
73. Write a Python program to display numbers from 1 to 10 using a for loop.
74. Write a Python program to read three numbers and find the largest among them.
75. Write a Python program using try-except to handle ValueError when a user enters a non-numeric value.
76. Write a Python program to add two numbers entered by the user.
77. Write a Python program to display all numbers divisible by 5 between 1 and 100.
78. Write a Python program using try-except to handle division by zero.
79. Write a Python function to calculate the area of a rectangle.
80. Write a Python program to convert a string into uppercase.
81. Write a Python program to find the square root of a number using the math library.
82. Write a Python program to generate a random number between 1 and 10.
83. Write a Python program to read a CSV file using Pandas.
84. Write a Python program to display the first 5 records of a CSV file using Pandas.
85. Write a Python program to write "Hello Student" into a file named "student.txt".
86. Write a Python program to draw a square using the Turtle library.
87. Write a Python function to calculate the volume of a room.
88. Write a Python program to print even numbers from 1 to 100.
89. Write a Python program to display a "Welcome" message using user input.
90. Write a Python program to find the greater of two numbers.
91. Write a Python program to display Pass or Fail based on inputted marks.
92. Write a simple Python program to ask for a user's name and age using input() and display a greeting.
93. Write a Python if-else statement that checks whether score is greater than or equal to 40 and prints "Pass" or "Fail".
94. Write a for loop to print the first five multiples of 3.
95. Create a user-defined function named multiply_numbers(x, y) that returns the product of two numbers.
96. Write code to import the math library and calculate pi multiplied by 10.
97. Write Python commands to open "records.txt" in append mode, write "New Data", and close the file.

98. Write a Turtle program to draw a straight line of 100 units, turn left 90 degrees, and draw another line of 50 units.
99. Given pandas imported as pd, write a one-line command to load "inventory.csv" into a variable named df.
100. Write a try-except block to catch ValueError when a user enters text instead of an integer.
101. Write Python code to import matplotlib.pyplot and generate a simple line plot using $x = [1, 2, 3]$ and $y = [10, 20, 30]$.
102. Analyze what happens when a while loop has no increment/decrement condition. How can it be fixed?
103. Write a Python function named area_circle(r) that returns the area of a circle using 3.14 or pi.
104. Explain how input validation and exception handling can reduce crashes caused by invalid user data.
105. Rewrite file-reading logic with try-except to handle FileNotFoundError gracefully.
106. How can exception handling prevent a program from terminating unexpectedly because of an error?
107. How can Turtle graphics make learning geometry more effective? Explain with reasons.
108. Write a Python program to draw a hexagon with Turtle using a for loop and calculate the turn angle dynamically.

UNIT 5: ARTIFICIAL INTELLIGENCE (AI) AND CONTEMPORARY TECHNOLOGIES

1. Write any two applications of Artificial Intelligence in daily life.
2. Explain the concept of Artificial Intelligence (AI) and Machine Learning (ML). How is ML a subset of AI?
3. What is supervised learning? Give a real-world example.
4. What is unsupervised learning? Give a real-world example.
5. Differentiate between supervised and unsupervised learning with practical examples.
6. A system predicts results using past records. Which type of machine learning would be suitable and why?
7. A supermarket groups customers according to buying behavior without predefined categories. Which machine learning technique is appropriate?
8. An online platform recommends products based on previous purchases. Explain how machine learning is involved.
9. What challenges may arise if machine learning models are trained on inaccurate data?
10. Why is testing important before deploying a machine learning model?
11. Why is bias considered a serious ethical problem in AI development?
12. What severe consequences could occur if bias is unintentionally programmed into an AI system used for diagnosing diseases in a hospital?
13. Why is data privacy a major concern when using Artificial Intelligence?
14. What ethical issues may arise when governments collect large amounts of citizen data?
15. Write any two rules or best practices that students should follow when using AI tools responsibly.

16. A student uses ChatGPT to write a history essay and submits it as their own work. What specific ethical issue does this raise in an educational setting?
17. Why can maintaining academic integrity be difficult in the era of Generative AI, and how should schools adapt?
18. How does Generative AI differ from a traditional internet search engine?
19. Name a chatbot that is useful for generating texts and images.
20. Analyze the impact of Generative AI deepfakes on social-media security and public trust.
21. Discuss the role of AI in robotics.
22. Define Robotics and mention fields that contribute to it.
23. How does Artificial Intelligence help robots in decision-making and safety?
24. Explain how AI can be used in a simple robotic task such as obstacle avoidance.
25. Define the Internet of Things (IoT) in simple words and give daily-life examples.
26. How do interconnected devices in IoT make a city a Smart City?
27. Why is internet connectivity important in IoT applications?
28. How can IoT technology be applied to modernize agriculture (Smart Agriculture) in Nepal?
29. How can IoT sensors and AI prediction models work together to detect and prevent forest fires?
30. How do smart cities use AI and IoT together to improve urban life?
31. A city wants to reduce energy wastage. How can AI and IoT be used together to achieve this goal?
32. How do smart parking systems use IoT technology?
33. How can cloud computing, AI, and IoT be integrated into a smart healthcare system?
34. Why do modern businesses prefer cloud computing over maintaining their own local servers?
35. Write two differences between traditional computing and cloud computing.
36. Why do businesses prefer cloud storage over traditional storage?
37. Compare cloud storage and local storage.
38. How can cloud computing help small businesses reduce costs?
39. How does cloud computing support online learning platforms?
40. How does cloud computing enable remote work?
41. How can cloud computing support collaboration among team members?
42. Explain the importance of data backup in cloud computing.
43. What problems might occur if a cloud service experiences downtime?
44. A company stores all its data on cloud servers instead of local computers. Discuss two advantages and one risk.
45. Explain the potential security risks if a government stores highly sensitive national-defense data on a public cloud server.
46. Explain the concept of SaaS (Software as a Service) and provide one common example.
47. A school uses Google Drive to store student records instead of hard drives. Which cloud service model (SaaS, PaaS, or IaaS) does this represent?
48. Explain why a software development company might find Platform as a Service (PaaS) more advantageous for building an app than Infrastructure as a Service (IaaS).

49. How do smart voice assistants such as Alexa or Siri use Natural Language Processing (NLP) to assist users?
50. Distinguish between Augmented Reality (AR) and Virtual Reality (VR).
51. Describe a scenario where Virtual Reality (VR) can be highly beneficial for medical training.
52. How can Extended Reality (XR) improve education and training?
53. State two benefits and two challenges of XR technologies.
54. Name two e-Education platforms (international or Nepal-based) and give one benefit of e-Education.
55. How can AI improve the learning experience of students in schools?
56. Write any two benefits of using AI in productivity applications for students.
57. Identify one e-Governance service currently used in Nepal and state how it benefits citizens.
58. Write two examples of e-Governance in Nepal.
59. A government launches an online citizen service portal. How does this support e-Governance?
60. How can digital technologies make government services more transparent?
61. How can the digital divide negatively affect the success and fairness of e-Governance initiatives?
62. What challenges might elderly citizens face when using digital government services?
63. A user purchases groceries online using a platform such as Daraz. Mention any two distinct benefits experienced through e-Commerce.
64. Contrast the cybersecurity measures needed to protect a global e-Commerce website with those needed for a traditional physical retail store.
65. Analyze the dual impact of AI on the job market. How can AI simultaneously create new employment opportunities while destroying traditional jobs?
66. Explain two positive and two negative impacts when a company replaces some human workers with AI systems.
67. Explain the potential negative effects of pure e-Education on a student's psychological and social development.
68. How can a country use AI, IoT, cloud computing, and e-Governance together to improve citizens' quality of life?